

Ryan Smith ~ Active Inference Insights 013 ~

Wellbeing, Modelling, Precision

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hello everyone and welcome back to active inference insights today I have the real pleasure of speaking to Ryan Smith Ryan is a principal investigator at The Iaua Institute for brain research and he's the research associate professor at oxy College of Health Sciences at the University of Tulsa his work focuses on the neurocomputational mechanisms underlying emotion cognition interactions and how they relate to brain body interactions with special focus on planning Information Gain and interceptive inference he is also the primary author of probably the best known and most highly rated tutorial on active inference writing it alongside KL Friston and Christopher White Ryan thank you for giving me your time uh welcome to the show thanks for uh thanks for having me excited to get Talk No it's it's an absolute pleasure it's it's really very exciting as I said I mean obviously I'd come across your work uh kind of inevitably for me to have invited you but it was actually a real treat to sort of go back and go through like what feels like a you've done loads and work at this incredible velocity um so it's been really fun to dive in and explore this I guess what felt like a more empirical side um because I'm spending most of my time reading sort of philosophy papers so it's great to see some like traditional statistics and some proper empirical work so so it's been a great it's been great fun on my part I wanted to start at kind of a pretty broad level and and speak about subjective well-being so you wrote this paper in 2022 sort of grounding subjective well-being in the parameters and mechanisms of active inference um you alongside Cas Beal of uh cesp have come up with this really nice way of grounding uh well-being in expected free energy but as you write in that paper you say um while variational free energy can be understood to track basic biological Fitness the this is neither necessary nor sufficient for an individual to report High subjective well-being so maybe where we could start for the audience is just explaining why the minimization of variational free energy is not sort of the simple solution to obtaining High subjective well-being sure yeah and I mean it's a definitely a very kind of complicated and yeah complex complex topic jump into and it honestly depends a little bit about the exact kind

of formalism that you're kind of working in I mean me I almost exclusively work within the um sort of like pomdp formalism um associated with decision-making and planning processes that uh that um you know as the focus for our tutorial and you know the focus of the tutorial was very much like trying to get people to a point of being able to do true kind of experimental empirical applications right I mean that's the final section is all kind of meant to transition into that um just because that's primarily what I do um so so I mean with respect to the kind of um yeah your question here is that within the p P scheme um you know variational free energy is primarily used as a means of optimizing posterior beliefs about States right the state inference um sort of process whereas expected free energy is the thing where you're trying to say okay what do I expect the kind of rewarding of the outcomes to be and how much information do I expect to gain you know conditional and choosing X versus y um so that there is a true kind of veency thing right in the expected free energy because you have this expected Divergence or lack thereof between the things you want and the things you expect right under a policy whereas the variational for energy is much more um a function of it's just a function of the observations you're getting and how you update beliefs about the most sort of appropriate approximate posterior over those States right so in other words you know if I perceive that uh you know the observation I'm get getting is that you know someone's like stabbing me or something right then the the process I go through to minimize variational for energy that that is what I use to infer that the best interpretation of what's happening to me is that I'm currently being stacked right so so if if the idea with variational free energy as it is at least in in the kind of standard PDP formalism um is a way of just getting at the best posterior belief about the state you're in then that state could be something you really like or something you really don't like right sure um so so just when you when you see it that way um then there is this kind of big distinction between trying to um trying to maximize um the consistency between your preferred observations and the observations you think you're going to get um that that's just that's veency and could relate to well-being um in a much more direct way than just trying to optimize your beliefs about the current states you're in right it's interesting that yeah you you focus more on the sort of perceptual side of variational free energy uh because at least the way way that I've spoken about it and I've seen written and I've and I've written is that it's kind of it's kind of dual wieldy in the sense that variational free energy being this uh kind of upper bound on self-information or surprisal you can also minimize it by taking action but then that said when we talk about action we always frame it within expected free energy terms so I've been kind of thinking just again this is just a completely sort of nent non really

fleshed out thought is that action seems to do can do variational free energy and expected free energy minimization at the same time in the sense that you know if you're uh sort of maximization of model evidence term the the minus log of $p(Y)$ if you're doing that and also you're you're maximizing pragmatic value it seems like you're doing both at the same time is it kind of a an erroneous way of thinking to alide or combine expected free energy and variational free energy in this way um yeah I mean like I said it's a little bit complex because I mean um so so okay we'll start put this um certainly um in the uh you know what the the accuracy of the expected free energy calculation that you do right depends right on whether you have good posterior beliefs Over States um and it depends on whether you're updating your actual generative model appropriately based on the patterns of observations you're getting over time um which would correspond to what we typically call learning right as opposed to perception um so obviously you're expected for energy estimates will not be good if your model is really bad um and so there's an indirect dependence right on our ability to maximize um being in the states that we want to be in um there's a dependence between that and using the um and minimizing variational free energy at least in a way that um um of course that depends on whether you're V minimizing variational free energy with respect to a model that's in fact accurate right I mean you can get an approximate posterior belief that minimizes variational for energy um with respect to a model but if your model is very mismatched with the world then right that approximate posterior is still not going to be a good description of the world and therefore you might still end up getting observations you really don't want when you choose policies based on that right so right um you know that being said I mean you know it's a little bit you formalize it because um the you know often when it's kind of talked about more generically especially in the philosophy literature um yeah definitely variational for energy has the kind of dual right perception action aspect but um but there there isn't really like a I mean the tricky thing is is that action at least when action is formed in the kind of explicit planning decision- making sense you still do require um some kind of cost function right you still have to have some mechanism that says I expect action a to be better than action B sure um and um and free energy is a function of current observations right whereas the the planning and decision making kind of has to be with respect to Future observations and that's kind of the distinction between variational and expected free energy um now now I say that in addition however I mean you so you mentioned the thing that me and Casper had done with u you know deeply felt affect paper and in and there there is this kind of further further term that can be added into the into the active inference formalism if

you want to it doesn't always kind of have to be there it depends on the you know context of the sort of task or cognitive process you you care about um or that you're trying to model but um where you you are effectively comparing the expected free energy to the variational free energy with the new observation that you get right and so there either will be a match or a mismatch there right so if it's the case for example and under with that example if it's the case that the um the expected for inges gets a little mathy but but the expected for energy Vector over over policies if that kind of geometrically if if that Vector points in a direction that is consistent with um the variational for energy of the observation that you get right then you can kind of think of that as being a little bit kind of like a positive reward prediction error it's basically saying that what you got was um consistent with and even sort of better than expected um compared to G so it's kind of like f matches with G or is consistent with G or maybe even a little better um nice and and that in the in the deeply felt affect formalism the idea was that that that uh sort of positive prediction that positive reward prediction ER effectively can itself be treated as an observation that can kind of feed into a higher level model where it's kind of like evidence for a a positive effective State and then that can kind of put a prior back down on what that what the kind of expected value is for um the parameter associated with that which is just this beta hyper prior thing um and you know the function the function of that that is interesting right because in the in the reverse Direction you can also have F disagree with G right which which would just mean that right you're expected free energy was based on a bad model essentially right um and so and so um you know in that case uh you would expect that there' be a a negative effective response um and so that would act act for act as evidence for this higher level model that would say okay I'm in a that's evidence for a negative effective state but what that um what that beta term ends up doing right so when you negative affective State corresponds to that beta value getting bigger um which ends up being um what it what it does is a little bit um is a little bit nuanced as well though because if you don't assume that there's any kind of initial habit like prior beliefs over policies then um effectively all the beta does is just add noise to Choice um you could think about it kind of like a random exploration term in um like in like in reinforcement learning but it's just kind of saying like trust your expected free energy less right there therefore select actions a little bit more randomly from your posterior distribution over policies but if you instead happen to have precise uh habits right so you've just the last 50 times you know you were in this situation you chose option two um then um and it's kind of assumed implicitly that that must have worked out well right if you kept doing it over and over again then then if beta gets big that it ends up

essentially favoring the influence of habits over um what the expected free energy would say um so in that case it kind of implies that uh when you're in a negative affective State you'll become more kind of habit driven um and so that's that's a little bit the way that that um the Dynamics end up working okay excellent yeah yeah yeah so maybe just myself and the audience I'm going to try and do a sort of paraphrasing slash just trying to align what what's going up what kind of what's going up on what's going down yeah so I'm familiar with the gamma parameter yeah the gamma parameter being sort of a scale of expected free energy estimates so it kind of tracks in a way how confident the system is in its action model that's the way that I think about it beta beta is just the hyperparameter on that it's just the inverse of that one over beta is gamma okay okay so that's so that's really useful so the way I thought about is that if you've got lower values of gamma that means the system has lower confidence in sort of its capacity to do expected free energy and so it's going to lean towards habits just so I'm aware that means that it's high uh beta yeah got okay so have you seen uh this new George Dean paper that came out the pre-print came out what last week something like this it's a wonderful paper it's about sort of canalized psychopathological behavior so very much in line with what you're writing about okay and exactly what they say so you can sort of treat gamma itself as a hidden state mhm and you can maximize your mental act you can optimize your mental action so as to kind of optimize gamma in a way so it kind of looks like okay I'm going to avoid thinking about me certain memories or I'm going to avoid thinking about expected you know a a potential future because that will lead I know that will lead to reduced gamma a lower estimate of my capacity to do expected free energy I may be butchering this um yeah I feel I feel I feel a little limited just because I haven't read the paper so I yeah I might not be able to comment confidently a l sure sure sure well I just wanted to mention it in any case because it's a it's a nice addition to this picture sure yeah and I mean for on our end yeah I can all I can really kind of say is is that yeah for for us or at least the way that we've been we thought about it previously is is is that gamma gamma itself is is it is not like sort of directly a hidden State itself but it is like an observation that canect as um it's it's a thing that can be treated as an observation to inform inference over hidden States at a higher level nice so you have kind of higher level that's has a posterior of AOK and a positive effective state or I'm in a negative effective State and the current and the the change in gamma after a new observation is one piece of evidence um for changing the the positive or negative effective State um two things that I think they're are kind of important there if if we're talking about it that way again that's just one idea but if we're talking about it that way one is is that crucially that kind of

implies that it's not the only thing that can act as evidence for the emotional state you're right so you could have lots of other things besides just this kind of positive or negative prediction ER thing that that influences right like I mean I would think that things about your interceptive state right like are probably important for that um um among among other things right so so I just I wouldn't I just personally wouldn't want to commit myself to saying that like it's gamma and only gamma is directly right this this thing um because the because in in that formalism what ends up happening is is just whatever your current belief is at the higher level about your affective State ends up putting a prior back on what gamma is right so like so when I'm in a negative effective State you know going forward on the next trial or the next time I have to make a choice that thing is pushing gamma down right um now I mean something something that I do probably want to highlight though is is is that I I doubt that that's the complete story um for for at least a couple of reasons one one big one is that empirically um negative affective States um while they are consistent consistent with what um consistent with what uh would be kind of suggested here negative affective States track something like lack of confidence in a model that being said um typically what that tends to look like in a lot of other sorts of modeling Frameworks and some empirical results is is that the main effect that that has isn't really to make decision- making noisier but actually to like turn up information seing and to turn up learning right right right it makes sense that if I don't think my model is very good then I ought to kind of turn up my learning rate so I update my beliefs faster um you know from the next sequence of events that I experience because I shouldn't be trusting my model well so it's kind of like okay I gotta hurry up and change my model to make it accurate again um and kind of influence of learning rate doesn't directly fall from anything about Gamma yeah yeah yeah yeah yeah so that's more I've been thinking about a little bit as well in terms of canalization that's more kind of this negotiation between the B Matrix and the a matrix um potentially might be one part of it right if I've got particularly imprecise uh B matrices let's say because I have some dopaminergic um synaptic misfiring and so I have hyper Precision over a then I'm going to mistrust these State transitions more than the norm let's say yeah I guess I guess the thing that probably matters here is is that um depending on the context that you're trying to model um there's very different things that you might need to be there's very different parts of your model that you might either want to be or not wanting to be updating right right so so so if so maybe right what I'm not confident in is something about my beliefs about State Transitions and in that case I want to be effectively or directly updating my learning rate about the transition probabilities right I should really pay

attention to the coincidences between State and next state over and over again to kind of like reoptimize that whereas if the thing that I'm um not confident about is something more about my likelihood function right more about the mapping between states observations um then I ought to be changing the kind of rate at which I changed my beliefs about that likelihood function um now you know how you do that itself is tricky like I said I think I think a more kind of extended formalism would need to be um put together if we were going to try to turn gamma into something that also affected that but um but it also I don't know one one kind of tricky thing I guess is is that um in a lot of the kind of practical modeling applications right now um the way that learning is um learning rate is adjusted is just via kind of fixed parameter um that's essentially just a a scaler on the um on the updates to the um to the the counts and the the counts that are added to the dally distributions um for you know a or b or you know what ever whatever one you're happen to you happen to be modeling learning at a given time um but it's a little bit I mean it it's similar to you know including these sorts of constants um the way you do in like a standard kind of reinforcement learning model um but it's not quite the same as um the way you might um do it if you were being kind of like trying to be fully true to basian learning um because because in in if if you were kind of going true Basi and learning completely and you didn't want to have a kind of like a hoc constant you know that just kind of scales the updates um then it it ought to be something that follows from the relative precisions of the um your prior and your um and the signal coming from the likel right yeah um and um you can do that um but and if you assume it kind of constant sensory input and a constant likelihood then you would learn more from that um signal given that you had flatter versus sharper prior of states which would follow from your transition beliefs and therefore you're B Matrix quote unquote but um that actually kind of when we've tried to do that in the past it works but it's hard to make it work without um making a model hierarchical or at least kind of making it kind of implicitly hierarchical yes it's a little tricky but it it formally the the reason for that is is that you um in the current schemes obser a are kind of treated as things that are um um kind of fully like one hot Vector like it's a you know a wanted a bunch of zeros basically no uncertainty about the observation where the uncertainty comes in it's just what state you should believe you're in given that observation but to to get the kind of learning that I'm talking about you need to actually have the the the obser to do it right in some ways you have to actually have the observation Vector itself have some noise yeah yeah yeah cross right and so that happens naturally in a hierarchical model because the the first level states do have noise in them and they act as observations for the second level I see but but also

you know for practical applications you can just kind of pretend your model is effectively hierarchical by just having it be single level but make the observations uh noisy as opposed to being one vectors and it does the same thing yeah well I was thinking it the other way that um to control these learning rates you would need something like a metacognitive aspect uh yeah you could do something like that too you could have like a higher level like a second level state that sort of conditional on that whatever that is it modulates the learning rate at the level below let me go back to this um this George Dean paper just because again I'm I'm quite concerned for some reason on well not for some reason it's for good intellectual honesty that I don't misspeak and I don't uh you know so and I'm sort of curious again about what you think so I've just got it up here they say um yeah so in effective inference the system not only possesses an estimate of action model confidence described as implicitly metacognitive but another layer of the generative model can be added so that the confidence estimate itself becomes a hidden state that the system infers using the same mechanics of perceptual inference so that's kind of similar to in a sense what I was saying right the the gammer itself is becoming the hidden state but I think what the point that they're making there is that you're not just doing perceptual updating you're also doing uh planning as inference so you can do in terms of mental action attentional schemes you can deploy certain mental actions such that the observations you make are going to be either you know increasing the likelihood of being in that hidden state or not and that's what you know that's whe whether that's thinking about something that's bad that's going to happen in the future or something bad that happened in the past you're more likely to avoid thinking those things because you're aware that probabilistically you might end up in low gamma yeah that's interesting I guess there's a few different things here where I'm just I don't know I mean my initial go-to and we start talking about all of this is kind of like how I would like you know write it down you know yeah yeah yeah make it kind of you know work formally and so there's kind of like a few different things that kind of mixed together in there um so I mean like one one thing I could totally see which I thought was what you were talking about a second ago was was like um instead of the gamma based kind of hierarchical updating thing that we were talking about before like that kind of second level State part of its likelihood back down to the first level could also be something about like the the learning rate right so it could directly through this kind of additional Connection in the second level likelihood you know when I'm in a negative effective State turn up the turn up the learning right at the lower level um which isn't so like so much intentional or kind of like controlled right it's just something that kind of like happens

automatically as a function of whatever you infer your effective state is um that's a little different I suppose from if you built in a policy itself that controlled something like the like transitions to like attentional States yes so we've done that we've done that some more like emotion inference models right where you can kind of you could kind of selectively choose okay now I'm going to pay attention to my current veilance all right now I'm going to pay attention to my arousal level um where you can kind of use a second hidden State factor to kind of like pull out slices of the likelihood tensor that that encode the Precision of the observations um um with respect to the mapping to a different hidden State Factor um but uh but so there's there's that kind of thing you don't actually need explicit hierarchy for that you just going to need separate orthogonal State factors like in practice but um but yeah so you could do that and then the third thing I guess it makes me think of is is something that I I played around with a little bit that I think is actually a really cool idea but haven't done anything really with is that um you can try to account for like what can still be like a kind of like basy in form of like motivated cognition that would look kind of biased from the outside by um you know because if in a second level model you're um you're treating the first level States as observations for the second level then that means like you can have a prior preference distribution over yes first level States right so then effectively like you actually just like want to believe some things and not others right you you want to believe some things and not others then like the optimal policy um will could be something that looks super anti-asian from a fully like epistemic perspective sure but probably is optimal under the actual scheme that you've got in place yeah yeah yeah exactly right but but externally right yes I mean this is exactly the point they're coming to they they speak of sort of avoidant mental action or motivated in attention right um and this idea is that you're kind of Performing the second order inference over the model precision as I said but you have this kind of predictive model with model precision as the hidden State such that I mean the way they talk about is how will I feel if I do that but with respect to mental action sure which I think I think it's very I don't know I just thought it was very cool and very much up your yeah absolutely I mean yeah when it comes to the affective stuff I totally agree like I said I'm super interested in this I mean it uh I mean in part of makes me like like I both think it's really cool and it also makes me slightly like empirically uncomfortable and the reason for that is is that like I'm pretty sure it makes like like actually testing like whether a model like this is kind of like the the best model of human behavior basically I think it makes it kind of UN fiable right right right right well it can account for basically anything but let's jump into that then I mean you when we actually spoke

on email you were Keen to sort of mention that you're not just doing pure simulation work and you're doing empir what you call well I think what is called empirical computational Psychiatry whereby you fit models to behavioral data in clinical populations which is super cool but I'm not a mathematician so and I assume a lot of the audience aren't mathematicians either so perhaps you could just in sort of a broad brush approach describe the distinction between uh pure simulation work and empirical computational Psychiatry yeah I mean I guess I think the the cool part about doing simulation stuff is is that you're like pretty under constrained right like like there's a there's a kind of like creativity about it and you can kind of get really complicated and you know it allows you to kind of simulate some cool mechanisms um but a lot of times like the the concern I suppose is that they become a lot of like Just So Stories in a sense right like because you know I could come up with probably like 10 different generative models that would uh you know that would hypothesize somewhat different mechanisms um but they would produce all the kind of like behavioral signatures of the sort of thing we're trying to explain right right so so I think it's important important to be able to find ways to actually you know test you know which of these models are actually good you know like accurate descriptions of what's actually going on in the brain um and um and so you know for that right you need to get these models to have direct contact with data somehow um and um and so usually you know so the sort of thing that you know we will do and and what I should say another thing is is that um you know like how good a model fit data um is always a relative comparison right I mean you can't there are absolute metrics of model fit but usually you want to kind of say okay here's a space of possible generative models um you know that that hypothesize different sorts of algorithms that are being used or or or different just model structures but that assume you know underlying the same underlying algorithm um and then you kind of want and you want to do like bayesian model comparison right just say out of all these models which model does the data provide the most evidence for um and you know like that that actually allows you to be able to test whether or not like an active inference model um or what an active inference model would a given active inference model would predict um you know actually reproduces behavior that people actually do right right um and um so so I just don't yeah I mean to be to be scientific about it I just wouldn't want or I think I think it's important that for people to take for for people outside of just the kind of like active inference Community proper right to actually kind of take these things seriously and think that they're moving um moving science forward um that uh you know we need to be able to actually show that these things are improving right how well we can predict um what brain and behavior

actually look like compared to lots of other simpler models that have been around for you know much longer than active inference right so so um I mean you know comp computational Psychiatry is a fairly um you it's fairly new field but you know there's a lot of there's more and more people working in it but most of them are not um coming from like an active inference background you know they more tend to focus on simpler reinforcement learning models um that um you know at the end of the day don't tend to be really too different in what they predict from from active inference in a lot of cases um so it you know the trick is kind of you know finding the right kinds of decision-making or Behavioral or perceptual tasks to have people to have you know participants come in and do whether they be healthy or whether they be patient groups um to try to um you know to see in other words you have to find tasks where it's clear that a particular active inference model would actually make different predictions than a simpler um reinforcement learning model and um you know we've started to do quite a bit more of that I am you know there's a you know for anyone for anyone who wants to come a little plug uh the um one of my uh current grad students um is uh presenting um well she just about to submit for a talk or a poster um for the computational Psychiatry conference um this summer in Minneapolis in July um and it's uh and this has been kind of our most thorough attempt at this um lately is we just have a a fairly simple like explore exploit um sort of like three arm banded task that involves trade-offs between information seeking and reward seeking and so it does kind of match right where where active inference is supposed to kind of show show its strengths with the directed exploration um in this case primarily with respect to the uh novelty term right so this would be like parameter exploration not State exploration um but um but so more or less we identified I think like this is like table essentially of um something like 30 different possible RL models and like 30 different equ like equivalent complexity active inference model something like is a really massive model space um and you know fit all these models to um to the um to behavioral data and a couple hundred people um and both a both a Taiwanese sample actually and an American sample um and um and just done model comparison across these um and um and the the models that tend to win are more complicated models um there's a the the two the two kind of best models in what we found so far one is one is an active inference model that assumes um that people have um one forgetting rate um and two separate learning rates for wins and losses um separately so basically update your beliefs more or less when you get a win versus when you get a loss um and then a kind of like Dynamic decision noise um sort of term that um is based on the the beta thing actually based on the based on the um updates of the uh the beta gamma um term that we were talking

about where where effectively what that ends up meaning is that um when people get an outcome that's unexpected right then gamma goes down and so the posterior policies flattens out and so essentially their choices become more exploratory in the in the randomness sense um when they get outcomes that they weren't expecting right if they were negative if they get outcomes that are better than expected and it's positive then they become sort of more deterministic in going with with what the model would say is the best choice um and so that's the best like active inference model that tends to fit so it's very complex it has like five or six parameters something like that um and the the winning RL model is slightly different it assumes Dynamic learning as opposed to Dynamic forgetting or I mean as opposed to Dynamic decision noise um but the the point is that the main the main difference between these models ends up not really so much being differences in exploratory Behavior so much as whether or not they include um basian forms of learning um right right the main thing that the main thing that active inference does that's really different when you match it to kind of an equivalent complexity RL model is um is that active inference models are kind of with the derish distributions again the details of that don't matter for non-mathematical people it just basically means that in an RL model you're just kind of updating your beliefs about the expected um reward probabilities whereas an active inference you're keeping track of those reward probabilities but also you're keeping track of your confidence in those beliefs about the reward probab abilities um and um and so that's what um the the kind of take-home with it for for what for what she's done is um mainly that the um the ability for the two models to um predict behavior is actually the best two models um is actually fairly similar but basian model comparison does um select the active inference model over the over the uh RL model um and because the differences in accuracy between the models aren't uh aren't amazingly different um ends up relating probably a little bit more to like complexity cost differences um and you know kind of evaluation of the independent like how independent each parameter is and explaining the behavior things like that um so so anyway I mean so a lot of what we've been doing is kind of attempting to say okay well you know how much better you know does active inference actually do right in explaining in explaining behavior behavior is it the kind of most parsimonious accurate explanation for um how people actually do make decisions in information seeking contexts um so that's a little bit you know where you know kind of the approach that we take so that and then using those things um by also having people in patient groups with like depression or anxiety or substance use disorders um do these same sorts of tasks and then see whether um you know we can explain clinically relevant differences in

these you know between patients and healthy comparisons um through differences in the actual parameter estimates right so maybe it's the case that so one thing that we have found right is is that pretty consistently substance users in active inference models tends to show um much slower learning rates from negative outcomes um than positive outcomes so they kind of it takes a lot more times of them choosing a suboptimal thing before they kind of change what they do um I found that I found that an interesting result uh your 2023 paper elevated decision on certainty and reduced avoidance drives in depression anxiety and substance use disorders I may that's a different one but yeah this might be slightly different but I found maybe maybe I misinterpreted what you said there but this idea that um individuals at least with substance used to S Show A reduced sensitivity to unpleasant stimuli yeah okay so so it's yeah so there's a yeah so that's a that's a with a different task um it's actually the same the same sample but it's a different task I guess the prior one is learning rates right the first one yeah so it kind of depends right the learning rate one again that's with like a bandit task and so it's it is about kind of EXP expl exploit updating beliefs about reward probabilities Etc whereas the approach avoidance conflict task is a little different it's not a learning task it's purely just kind of a planning forward-looking task yeah yeah and in that one it's more or less there's a different kind of number of you know points that you know people can win but they have to also choose to go through some pretty unpleasant aect seeing some pretty seeing and hearing some pretty unpleasant effective stimul at the same time so you going to have to plan whether okay do I prefer to kind of move toward the side of this Runway that means the probability is higher I'll kind of get this safe neutral outcome but no points or whether I'll whether I expect it to be kind of worth it to me right to like under a certain point value try to act in a way that will increase the probability that I'll get the points but also see the super unpleasant stuff right and and in that case there's um you know this is sort of task that's been used in non-computational modeling ways several times in the past but the the main kind of metric if you don't do it the computational way is more or less just this General kind of unidimensional approach avoidance thing whereas the the modeling version is a little bit more helpful because we can estimate the beta parameter for these people and then we can also estimate um certain things about the shape of their preference distribution um and um and what that shows so we can kind of use that to dissociate two different aspects of approach avoidance conflict one is this kind of like uncertainty about decisions and the other is this actual kind of you know approach avoidance drive right um and the yeah the results for that are actually kind of surprising you know because like in the in the depression anxiety and in the substance users um they tend to

show um greater the patients tend to show greater decision uncertainty so they have higher beta values but they um they actually show um more precise preferences for the for the points which basically means that they're means that they're kind of more willing to go through the negative AFF stuff I was very surprised by that yeah so they're actually less avoidant right so they're more uncertain but they're less avoidant and that's we've now shown that in four studies you know so it's a very consistent result across multiple samples I read that and I had to read it twice uh because I was really like I really expected the other way around because we're used to hearing that avoidance is the kind of number one driver of a psychological disorder like generalized anxiety disorder or so what what do you imagine is going on here I mean I think one just talking in terms of very lame and non computational terms yeah there's there's the overexposure of people with psychological disorders to negative stimuli in the first place so what's negative for a healthy control is well it's all relative right and if you've been exposed to overtly negative things like your own emotions over and over again maybe you have a reduced sensitivity yeah I mean there's you know I mean we don't we don't really have like unfortunately I mean our our hypothesis going in was absolutely what you said originally right that we would get more avoidance in the patient groups um but uh you know we were just hoping that the the modeling would help us kind of distinguish these couple different kind of Dimensions right of of of approach avoidance conflict um but uh but definitely we didn't expect the the more approach stuff in the in the in the patient groups um that being said yeah so we so because of that we don't have kind of a bunch of other we don't have all the measures that we probably would have gotten um to try to dissociate different possible explanations for that kind of unexpected avoidance uh or unexpected lack of avoidance um so it it becomes a little speculative but um one thing is absolutely I mean that's one thing you might you might suspect is what you said that um you know if if people uh if people just in general are in this kind of heightened negative affective State then you know you might expect that makes them more sensitive to other negative affective things but you also might make that also might think they're a little bit habituated to it yes exactly what would that look like in terms of a kind of PM DP scheme because I guess we associate habits with action with the priv value over or priv beliefs over action when we're talking about habituated uh responses to stimuli I mean I guess it is it's in a sense action but what what shape does that take um yeah I mean so so like I said in in the official model that we were fitting it doesn't really include any include a the Dynamics of habituation we're just kind of estimating the what the shape of their preference distribution is right to explain their behavior um and so it so then the explanation just ends up

being that they have a more precise preference for for winning the points or or or less the the the the magnitude of the the probability of the negative outcomes in the preference distribution which again doesn't really mean probability it just means they they dis prefer the negative stuff less right um is uh is um you know that ends up explaining it but but um but what would it look like if you included some sort of actual learning dynamic for um for habituation um I mean so this kind of habituation is is really not so much more about habitual action as it is just the level of sensitivity you have to stimulate right so yes so it's a little more has this flavor of okay if I see something over and over again then my response to it each time is less right so right so um I think you could cost that in terms of action um it it well I mean it's okay so I'm thinking in terms of traditional therapeutic responses right like in terms of exposure response prevention therapy for example you would you at least have an element of it which said that a uh dysfunctional avoidance response again this is a non-active inference way of framing it but just in a Layman sense it entrenches the the the severity or the perception of the severity of the stimulus yeah so I mean we have you know I mean I wouldn't necessarily have expected you seen it but happen to I mean we do have this paper we published a couple years ago as a simulation paper specifically on mechanisms of uh possible mechanisms of CBT in in a pal DP so we actually did simulate exposure therapy in particular nice um and um but it was The crucial kind of thing we were looking for there was whether or not we could kind of capture um make predictions about the ways that cognitive interventions like reappraisal would interact with behavioral interventions like exposure therapy um and um and I mean some of some of some of that we ended up being kind of kind of excited about it seemed consistent with results or with you know when I talk to clinicians because I work with a lot of clinicians um you know what they say but where where with that I mean the the kind of you know habituation was much more about like so if a person has a strong if a person has a strong expectation that like if they choose a certain action then the things they are going to observe are things that they really don't like right so things that have a very low uh preference level um then you know it follows all us being equal they're not going to they're going to take actions that don't lead them to those to make those observations right but the problem is is that when those when those um when those uh uh beliefs about the expected outcomes of actions are are wrong right then if they just avoid it every time right then they can they can never update their beliefs right so I mean exposure is essentially kind of this kind of like uh clinician facilitated directed exploration right right right you know they kind of push you to go take this action right that's gonna make you see okay what actually happens when I'm in this state that I've

been avoiding um and so and so the kind of habituation in that sess just ends up being once they start actually seeing What observations they really get when they're avoiding then they just you know it just updates their likelihood function right under that state um but um but the the interesting thing at least in in that paper was a lot more of how that interacts with um your um cognitive beliefs so because because you can also and this is often what happens right is is that when when people go through exposure therapy um they will they can um even after those respon the negative responses kind of go away and they start to say okay now no I believe like in the context of spider phobia for example like no okay actually you know I believe this spider is safe now right and so their responses their affective responses like died down but you know you bring them back in a couple days later they see the spider and they have the negative responses again sure right so it doesn't kind of like stick around it doesn't generalize um you know and so why is that right because under a simple under a simple kind of like unlearning story um they would have just overwritten their old likelihood function right so they shouldn't right their generative models just be different they shouldn't have those responses again um you know so what explains that well what explains that has to be that they um that they don't take evidence they don't take the evidence from The Unexpected outcomes when they see a spider um as evidence that their likelihood function was wrong they take it instead as evidence that they've actually transitioned into a new hidden hidden State context so that now so now they believe okay I am in the safe context and then they just update their likelihood under the safe context Ah that's interesting right so then all you have to do is say okay well if there's any uncertainty in the transition function then even though they believe they're in the safe context now to come back in a couple days later there's more uncertainty over what context they're in so the responses come back because the likelihood function under the danger context still says right bad things will happen if spider so how would you resolve that um yeah so so it um I mean so one thing you want to do then is you want to you want to try your best as a as a therapist in some way to um to not use cognitive approaches to try to like make people believe oh actually this is really you know this is really safe right yeah you're gonna go in this room you're gonna see this spider yeah but it is safe right right like you have to have them actually maintain some kind of uncertainty about what context there in in right because only in that case if there's like a like in the simulations more or less like we showed is like if you keep the posterior distribution that they have about the context there and if you kind of keep that impr precise right if you maintain the kind of cognitive uncertainty then when they get the unexpected observations that bad things don't happen then it does overwrite the

likelihood under both States right um under both hidden both hidden contexts and then it does generalize right um and that's um that's actually uh consistent with this kind of cool study that um Sam gersman um has written some stuff about um back in like 2015 is it's like been a little while but but where you can do more or less the same thing with rats um with like just classical conditioning experiments where you know this is kind of you know you play a tone they get shocked you know like over and over again with some probability so now you play the tone and they kind of jump or freeze or do something implying that they'll shock um and then the standard thing to do during Extinction is you just present the tone over and over again you don't show the shock at all or you don't give shock at all and so then their responses slowly die um but similar to with exposure therapy you bring the rats back a day later you play the tone again and they do have this kind of spontaneous recovery is what it's called where they start to have the fear response again um which means they're doing basically the same thing right the rats too are just inferring okay I'm not sure if I'm still in the safe context um so so what they showed though is that if you do the extinction um if you do the extinction time course a little differently um then you can actually prevent this spontaneous recovery um and the way they did it is instead of during Extinction just abruptly completely stopping the the association between the tone and the shock what they did is they just said okay for a little while tone is assoc associated with shock 90% of the time okay now for a while tone is associated with shock 80% of the time okay now 70% of the time they tape of them off yeah so and what that does you can kind of think about it is the prediction errors are never that big right and if the prediction errors are never that big then that's not good evidence for the shift in context right right so does cause this slow overwriting right of the original likelihood function as opposed to promoting the state inference thing and when they did it that way and then they tested their routes later they didn't have the spontaneous recovery like the the fear responses didn't come back um and that is fascinating it's very it's it's quite counterintuitive though yeah I think you would one would be under the presumption that if you just kept providing evidence you know if you kept tying an observation to a different state that would just become entrenched but it it does make sense that that itself becomes this kind of crystallized thing and then once you take them out and put them back in then the previous one still has that Precision that's that's that's really fascinating you also said something very interesting there which was um it was kind of under the radar but the difference between having a strong preference to be somewhere or to have something or to be something or whatever versus uh H like having a strong dispreference not to be somewhere be something have something so a positive versus a

double negative M I've often found this in in active inference that I don't think it's been particularly well codified what the difference is between those so people say oh um I want to be um 37.8 right in some subpersonal way I want to be 37.8 degrees Celsius and so I take action to restore that but you could well cast as I don't want to be 40 degrees celsius yeah right is there a is there a technical difference between those um so I mean one one thing that I guess is a is kind of unique about the way that active inference en codes preferences compared to you know more kind of like traditional formalisms like reinforcement learning that cast things explicitly in terms of reward function um is that you know strictly speaking an active inference agent isn't really trying to maximize reward they're trying to minimize Divergence from a target distribution right and that actually has certain interesting implications um but one one uh one thing to consider with that though is that is that um unlike in things like reinforcement learning um all of your preferences kind of constrain each other in the sense that the whole distribution has to add up to one right um so so uh you can um so I think that you will have to kind of relatively speaking um basically you will only be able to specify relative preferences um so so I think that so I think that if you if you have a distrib preference distribution where the kind of highest mass is over you know like 38 is degrees right then then the fact that it kind of tails off on both sides simultaneously guarantees that you dis prefer being above or below that um so so I don't know explicitly whether or not there is um whether or not there's there's any way to um encode those in a way that's different um there are interesting things you can do by putting kind of like negative preferences for something and positive preferences for another thing if there's like an observation you can lock as kind of like the Baseline value of like zero um when the thing gets normalized um to turn it back into an actual probability distribution then then it does still end up being kind of a precise uh preference for one thing and dis preference for the other and they do still end up being relative to each other though so um and the main the main thing that we tend to see when we do like simulation stuff and those sorts of settings is it just tends to um lead to this kind of like anti-exploration behavior um when when you know one thing's really negative where it's kind of like the uh the uh agent is just kind of this like you know better safe than sorry you know sort of conservative behavior um but I um but I think there are there should definitely be ways of encoding those in terms of all just kind of relative relative positive values that end up normalizing to the same distribution cool yeah I I think part of the reason why I asked that is because I think it's a point that you making uh the papers that you've written on interceptive inference which is that when uh we speak of sort of So-Cal aberant inter receptive inference a

lot of papers are not very clear whether they're talking about the underweighting of Prior or the overweighting of the likelihood distribution sure um I think part of the problem is people think of uh Precision weighting as like a necessary Zero Sum game such that you can't have both so if I have uh reduced Precision over my likelihood by necessity that will lead to increased Precision over my prior if I decreased Precision of my prior by necessity that leads to an uptake in the Precision of the likelihood but what you show at least in this 2020 paper the one that I read from plus um where you did the interceptive perturbation condition is that this isn't the case so where where do you think that kind of false intuition comes from that oh well it comes it comes from it comes from the fact that people usually in the way that they at least in these more kind of conceptual papers um comes in part from the fact that uh these are Under um uh they're talking about models that have gaussian assumptions with continuous distribution um because if you have if you just have along a single Dimension right if you have like a distribution that defines the likelihood function and a distribution that defines the um prior belief then then there are right when I turn up one or turn down the other then there's a bunch of different ways that I can kind of combine those things to lead to the same posterior um so it does become hard to distinguish um so it depends on the dimensionality and depends on whether you're were talking about continuous or categorical distributions um so so in in the in the case of what we're doing these are discrete distributions right so in discrete models with the dimensionality the way that we have it set up um it makes uh it makes clearly distinct behavioral predictions but um but in the uh in the kind of unidimensional gaussian um setup um then you you do get this more um only the relative weighting matters yeah yeah yeah yeah yeah that makes sense okay so thus far I I know we we haven't got too much time but thus far I know we've kind of been speaking generally about perceptual updating and parameter updating um or parameter learning and belief updating but we also have structure learning and active inference and I think you make a really nice point to say that sort of something like introducing a concept to distinguish between um or or to yes to let's say split up a concept that you have before so if I had anger now maybe I can turn that into rage and fury sure um I think I believe you call that emotional granularity yeah in the in the yeah in the emotion in the emotional state inference context yeah we would talk about that in terms of granularity or yeah that's usually the probably the best way to talk about it yeah it's a really fascinating concept because it it I think on on first glance it seems like a it seems like a sort of just linear benefit the more emotional gr granularity I have the better um yeah I mean what the what the direct relationship would be between like the uh the number of you know States in a state space and

uh and something to do with gamma I think is is going to be fairly indirect um but for a number of reasons but um I mean I think with most things right I mean the general answer is going to be that um it's not so much about complexity per se as more complexity than necessary yeah right so so um right you you know complexity adding complexity is good if it increases accuracy right so so um so what it would mean right to be more complex than necessary would be that there's a mismatch to the actual generative process so I think that I think that the um if you have more right if you're overfitting right to the to the generative process right if you if you're inventing more categories than there actually are distinctions to make um then obviously it's going to be problematic right because you're you're going to think you have more information than actually you do um but but so long as but so long as you're carving up your state space in a more fine grained way and that actually does add to the informativeness of the information that your generative model uh is able to kind of pick up and then use to inform policy selection then then I would expect it would be the more the better right it's just the more so it's the the more accurate information you have the more informed policy selection will be right so so as long as as long as you have you're good at inferring the right you know posterior over what emotional state you're in and each emotional state that you kind of have have acquired into your generative model um if if each of those does in fact make unique um um predictions about patterns of current and future observations under that state then then yeah I mean it'll be better just because it'll should right on average just make policy selection more informed right right right but um yeah thinking more broadly about the distinction between structure learning and parameter learning do you see these as bright dividing lines or do you see them as being kind of philosophically vague as to when parameter learning becomes structure learning um you know like I guess it's just it's just tricky I mean like you get into the weeds with some of this stuff because so you do that's what we're here people use structural learning the term structural learning to mean several different things right so I mean structural learning itself is is a kind of like superordinate category with a bunch of different kind of Exemplar things within it um so you know I mean we did I don't know if you saw this this just one paper on structural learning specifically that you know we put out a few years ago where it was kind of like you know people that you know there was just like kind of a dumb example of uh people trying to like learn like coarse grain or fine grain different types of like birds and fish and stuff why right yeah and and you know so that it was like you could just learn like these broad categories of Bird and Fish Or you could kind of like learn to carve it up into specific types of birds specific types of fish and they just kind of each of those was

defined with respect to specific like patterns of covariance among certain sorts of observable features right like wings versus gills and size and color and things like that but um you know and in that particular case the way that like structure learning was post was in relation to kind of like inferring how many states need to be in your state space right to account for account for data um and so and so um that has to do with kind of the setup of the structure of the generative model proper right like how many how many types of States how many states how many how many how many state factors how many levels of States within each state Factor Etc um you know so in the in the general pomdp scheme that is distinguishable from parameter learning in the sense that parameters are usually kind of um the the values of the entries into the matrices within you know things like the likelihood function transition function Etc um that you update but you update those under the assumption that like you know number of number of columns in the in each Matrix are are kind of fixed right right right so so if you're talking about kind of this granularity kind of fine graining of of your of the Concepts that you have um then then there is a clear distinction between um between uh those examples of parameter learning and structure learning but um like I said there's other types of structural learning and the actual implementation of how you do structural learning also matters because you know in our example um you know we we use this kind of like little trick right where we just kind of said you know we're we're actually gonna you know stick the matrices in as though there's a bunch of these kind of extra unused columns you know so we didn't have to kind of like explicitly add states to the state space we just kind of had a bunch of states that were just kind of like sitting there silently unused yeah and then the way that and then the way that the thing kind of worked was just that you know what they meant to be unused was the distributions were just completely flat right so they didn't predict really anything unique about the observations and so and so the way that that would work is just that um more or less that if you got some pattern of observations that was consistent with one of the kind of informative likelihood functions under a kind of state that you already had um then the posterior would be precise over that state and you know you would have a basically zero posterior over all the unused columns and so you wouldn't have any parameter updating under those columns but um but if you had some pattern of observations and they were really inconsistent with any of the informative mappings that you had then effectively what that would do is it would mean that the flat mappings would would better explain the data right right right and you know and as soon as like the flat mapping better explains the data then you end up with a precise posterior over one of these unused columns right whichever

one happens to explain it best and then you just start learning the parameters under that new column right so so and then you that's just a way of learning a new informative mapping so it's kind of like effectively now you have a new column you've expanded your state space right yeah and you could you could do that a little bit more explicitly than we did by writing in some additional function that like explicitly you know just like added new columns to the state space or something as opposed to just having these unused ones sitting around but my point is is that the the relationship between structure learning and parameter learning is very they're very tightly intertwined in in that case right because like when you start learning the parameters like that the thing that leads you to start learning parameters is the the exact thing that is causing you to do structure learning why why why why right yes exactly so again one that's one implementation right so I'm saying that that won't be true that doesn't need to be true um generically but that will be true in cases yeah yeah yeah yeah well and it seems like they all to yeah they all lean on one another in some kind of recursive manner where you can't really have structure Learning Without learning the parameters in you know of that structure in the first place nor the belief updating that allows you to kind of get there in the first place right in a dynamic system in a dynamically changing system yeah I mean I think I think it's just and again I can't I can't really say with confidence how well this will generalize to other cases of structure learning or or even just other kind of generative model setups and things like that but but yeah at least in this case it's kind of you you have to you have to effectively infer that that the current informative parameters in parameter based mappings that you have are not doing a good job of explaining something and then end up with a very low posterior distribution um over or a posterior that assigns a very low probability to any of the States you currently have in your state space and then yeah and then so it's like you get the right posterior based on you know that based on the fact that your current parameters are not good expl you know don't provide good explanations for the data and then by starting to update different set of parameters you end up leading to doing something that is effectively structural learning so it's yeah they're just very yeah it is all just very kind of tightly connected yeah yeah yeah well it'd be interesting to see if you could sort of disambiguate them you know in a model like whether you could just have I guess do yeah conceptually though I think that's tricky right because you somehow have to say I need structural learning only because well exactly that's why I meant kind of bu this recursive nature I mean I don't know if recursion is even the right way of talking about it it might be sort of nesting um in some sense that belief updating drives paramet updating which like ultimately feeds into structure learning yeah yeah cool so final

question big question you got this paper coming out the empir the so-called empirical status of predictive coding H and active inference but yeah and active inference I yes is that the is that the whole name of the paper yeah um it's a sort of classic critique leverage that active inference that it's self-contained and it's an echo chamber and it's a lot of philosophers and mathematicians kind of spitballing these fancy ideas but it doesn't this is a common critique I'm not saying it's true oh okay yeah yeah yeah yeah yeah oh well it would be it would be odd if it if I would to formulate that position as the host of this podcast oh no no no sorry I thought I thought you were saying that that was the the thrust of our paper no no no no no I'm saying it's a common critique from those on the outside I don't remember saying all of that no no no no I I know you didn't but it's a common critique at least I mean that might be uh oversimplifying the critique but that you know this Theory of Everything um is overgeneralizing and it doesn't have evidential an evidential basis where having done this all of Deep dive into the empirical status of active inference and predictive coding what are the conclusions that you guys have drawn um I mean I I just think I kind of think at the end of the day I mean it mounts to be something kind of similar to what I think you'll see in a lot of review papers for most things right is this is just that there's a lot of stuff where more work needs to be done sure to be for it to be definitive I mean that's kind of like the the very oversimplified conclusion I mean I would say that I I mean one thing that that I hope that we tried to make clear um um is just it's more that like there's just lots of different levels of claims with respect to both predictive coding and active inference and they all kind of need to be tested in a way that's a little bit at least partially independent of each other right so I mean especially in the philosophy literature people will even talk about really broad like categories like you know just like predictive processing you know quote unquote but like I mean if you want to move from that down to stuff that is actually testable um requires a bunch of there's a bunch of kind of layers that you have to kind of work through to get there right so you can have to say Okay well predictive processing it's very generic idea that in some way one of the main things the brain does is predict stuff I guess right it's like okay well you know two of the most kind of like prominent but certainly not the only possible categories for that are like predictive coding and perception and active inference and action but then even from there right I mean like predictive coding makes particular um explain or makes particular predictions if what you're talking about is um something about the uh the the the the predictive coding algorithm right as defined right which is which doesn't necessarily need to make predictions at the level of the kind of like neur neurobiological implementation right so and same thing same

thing with active inference right active inference for say okay well what type of active inference are we talking about are we talking about like the you know continuous State spaces are we talking about discrete State spaces and you know like are we talking about um you know commitments related to you know some of these like claims about you know the gamma stuff and dopamine or you know are we just talking about things about categorical State inference and that and that that depends on minimizing variational free energy um through like a variational message passing scheme right and okay are we actually hypothesizing variational message passing are we hypothesizing marginal message passing or are we hypothesizing that the implementation draws or that the algorithm is doing some sort of like belief propagation thing right and and then even there right that's still all just at the algorithmic level then you have to say Okay well if I commit to some algorithm right if I commit to PDP discrete State space you know this particular generative model structure for this task um you know with this you know with this message passing scheme you know as I said then all right there's like probably at least like 20 ways that you could set up a neural network right to solve that right so then you also have to have a hypothesis about okay what am I what what implementation you know am I am I testing right and what are the predictions of that implementation right so if I'm doing something at the Neuroscience level then I'm G say okay it's kind of this very kind of hierarchical thing where it's like you know this particular neural prediction that falls from this imple hypothesis and implementation that falls from this specific narrow algorithmic commitment you know like from this larger kind of framework you know what I'm saying so it's like you start making exclusions yeah so testing so testing these things is what it even means to be testing these things is is kind of complicated because um by the time you can actually get down to something that's empirically testable um you've narrowed it down a ton right and ruling out one implementation doesn't falsify yeah itive processing nor does it falsify active inference right it just means you ruled out that a particular implementation of a particular algorithm you know within the active inference framework Etc so it's like okay well now on testing like implementation hypothesis too so it's a you know so that that's tough right so then um but a lot of you know but a lot of what's been done is much more um well I should say with with predictive coding a lot of stuff's been done at the at the neural level with active inference it's very minimal has been done done in terms like Neuroscience um neuro neuroscientific studies like Philip shorton be had a cool study in like 2015 where they did um where they did uh some of the um some of the uh like midbrain dopamine um results in relation to um some of the uh the beta updating stuff I mean it's called Alpha in that paper so

like slightly older firm but but same basic idea um but so you know so there's that and then you know there's kind of like with predictive coding for example um you know I think I think many people would say that one you know one specific commitment at least to kind of like vanilla um forms of you know hypothesis predictive coding architectures is like okay well prediction errors get passed upward predictions get passed downward you know there's kind of these separate populations of prediction units and prediction error units yeah typically it's like okay well the prediction error units are in like cortical R 23 and you know like and the Deep you know prediction units are in you know parameter cells in layer five five and six um you know but then you know when you actually look at like some of these more complicated like canonical microcircuit models and things like that that that really explore in detail right how you would do predictive coding um in a way that would kind of match up to the actual like cortical columnar architectures that we know about you know from like micro anatomical you know actual brain studies like you know I mean it becomes way more complicated right you have like specific roles of like granular cells and then you have kind of like the sorts of predictions and prediction that are hypothesized to be like passed around through like different inter neurons and then how those pass signals up and you know turns out that like like uh prediction error signals need to be inhibitory but the between column connections are all glutamatergic and so they're excitatory right so you need to basically have any prediction are signal be excitatory but then pass through some inhibitory inner neuron at the receiving end to then turn it into right an inhibitory thing and then you know as opposed to the simple kind of vanilla PR predictive coding architectures or or algorithm um the most of the canonical microcircuit stuff has been pitched in terms of these um more kind of like temporally extended versions of predictive coding where you have like at each level you're not just representing a hidden state but you're also representing what they call like a hidden cause um and the the distinction between hidden State and a hidden causes is that like one of them actually predicts the Dynamics of the states at the level below um so it's kind of like hidden state to say like hidden state is or now I'm mixing up which one it is one of them either hidden state or hidden cause uh I mean this is just very jargony just arbitrary choice of terminology to a certain extent but but well I'll say it the way that's a little more intuitive to me but but so if it's if hidden state is like birds singing a song right at one level then you know I just have some representation about you know probability of bird singing song or something like that fairly abstract but then that then sends a signal to represent this hidden cause thing and what the hidden cause thing does is send predictions down to the level below about like some series of

notes right like in some sequence to the level where the hidden State at that level is like representing what note is happening at a unit you know some at some time period but so so even there right those are both predictive coding but those are two different predictive yeah different predictive coding theories um so so anyway I mean so just just to point out fairly longwinded that there's just a lot of complexity in in specifying what you're in fact even testing um but the um but the main the main thing I mean the main kind of broad Strokes result was just that um in predictive coding I mean there's certainly nothing that rules it out right I mean there's um there's some cool you know results Pro probably like the strongest results that um would suggest something like that has to be going on is um like findings related to like emission responses right where brain activity increases when you expect to get a stimulus and the stimulus is absent yeah right so like if a stimulus is stimulus comes right it should trigger some activity under a feedforward sort of hypothesis but the absence of something should never trigger yeah yeah classic shorts and monu yeah but um but the but emission responses and you know other things like error related negativity and a lot of the the kind of more standard stuff that looks like prediction and prediction error responses that are consistent with predictive coding um you know they certainly don't you know provide really precise evidence so there's like you know particular Layer Two Three neurons that are representing prediction errors and five six ones that are going to pred there are a couple cool studies that have come out like super recently though that do look like with like laminar fmri and things like that that do are starting to support these kind of like distinct populations of error neurons and pric neurons so that stuff's exciting um very well you know it's a niche but at least it's it's incorporating well at least it sounds like from your investigation it's beginning to well I'd like to think that it's beginning to down on some specific ways that we can get more precise hypotheses and ways of testing those hypotheses I mean it's it's it's it's read you know if the you know fre the the main papers 2009 2010 fair enough that it's taken a little while to get its you know it it to get to start reaching that kind of finer Precision that that that level of granularity that level of I think you know tomorrow some of that philosophical work needed to be done just to lay out exactly the parameters if you pardon the P of the expandom of what we're trying to find out so I don't know it sounds promising to me sounds exciting yeah no I mean I think yeah I mean certainly I think I think we need to hold things to a given I mean as you say right given that these um you know these particular theories are you know much more recent right than other like you know computational Neuroscience theories that have had a lot more time right to be to be tested um so so yeah I mean I think we need to be fair given the

amount of time that that um you know these things have been uh these things have even been been around to test and the you know the number of groups that have been motivated to you know put together studies and get them funded right to do these sorts of tests um absolutely um you know I mean and and with um and with respect to uh you know active inference I mean that's even more true right I mean the the actual kind of like empirically testable like behavioral modeling you know sorts of you know sufficient sufficiently precise right formalisms for um you know like the kind of stuffs that we do you know what's that's been around like what since like 2017 is and and it's you know been kind of evolving from there and and so um and so you know much much less time even and um you know I mean I think there's a lot of things that you know I mean we still need to do I mean along the approach that we're taking right so I mean primarily you know what we've you know the stuff that we review that you know that we've done with the you know empir like trying to find empirical support for active inference um probably hasn't really up until now done a good enough job of doing model comparison against competing models you know our Focus has been largely um kind of like computational Psychiatry focused so the idea has been more okay let's define a set of active inference models fit those models to behavior and see where we can find differences in those models between clinical groups and healthy groups right so we can show that like in an absolute sense right like active INF models could explain data pre you know reasonably well right like on a on a model on a task where like chance action probability would be like oneir you know like these models can explain behavior with you know assigning probabilities around like you know between like 0.65 and 0.7 something right so I mean it's it's you know much much better than chance and not uh not too different than what you'd see in in other um computational Neuroscience studies um but you know that hasn't that doesn't directly compare active inference models um to you know other classes of models right that have already been around right and so that's a little bit more the kind of thing we've been trying to do recently is to more definitively kind of say you know how much you know can we find evidence for active inference models that is not equal evidence for other models that have been around for longer yeah yeah yeah no that that it sounds absolutely worthwhile and yeah I think I think some people are you know they don't they almost don't want to do that because if it turns out that an RL model does better than an active inference model then like let's just throw in the towel but um yeah even even that I think needs to be kind of like understood with a little bit of a grain of salt though because because um well you can you can worry that that um that there are tasks where the tasks are simple enough that an RL model will do a perfectly good job

explaining what's going on and an active inference model would basically predict the exact same thing right right as a reinforcement learning model and in that case you know maybe the active inference model is more complicated and so like but you know I mean that's just because in those contexts they're you know they have a certain isomorphism you know to them so I think I think you have to you have to make these comparisons in tasks where um like a simpler model certainly could explain behavior equally well but but there's there's reason to think other models that that there could be the um patterns of Dynamics in the in the choice behavior that um that could differentiate right active inference models from reinfor from reinfor I like it Ryan I like it you're staying true to the to the field I like it it's good that loyalty is much needed but no I I I really do admire the the intellectual rigor and the academic rigor it's um it's much needed uh I wouldn't certainly wouldn't want to like say you know like our stuff and only our stuff I mean there are other people that are doing you know very recently that have been doing really cool stuff like there's this um like the first author last like gson or something like that I can't remember Sam gson Maybe I I don't quote me on that but but they did this really cool um um study where they took like these two um a couple I remember it was like four I think it was yeah four different like publicly available data sets um from uh on the two-step task um twostep tap is a very kind of classic um RL task that like basically was designed specifically to estimate how much people are doing more kind of like modelbased versus model free um sorts of planning um and um and I mean Super widely used I mean people have shown really cool stuff with it you know like like developmentally you know like kids as they get older start to look more model based you know right right right you know basically means they actually start to use an actual transition function more right um um and um but um but you know so what they did is they they took right like the classic you know widely used um modelbased model-free kind of combo model that typically gets used for um um for that task and then they also fit that same data to um to uh um like a competing active inference model um and um in two of the four data sets um model comparison was kind of equivocal um and in the other two data sets active inference models did win in model comparison um for that and these are pretty large data sets and again that's a very it's another you know very kind of classic task so the fact that uh the fact that active inference did a better job of explaining two-step tax Behavior I thought you know I thought that was actually like incredibly yeah no well yeah I mean when it starts better than what we've done in terms of actually testing active inference models yeah well when starts entering into the space where others should dominate that's you know that's when you know it's onto something Ryan I

know you're you're you're the world's busiest man uh filling out forms and and all sorts uh and so I'm incredibly appreciative we've just hit the hour and a half mark um well thank you so much for for giving me your time and educating me I feel I mean every time I have people on whether they're philosophers neuroscientists mathematicians and so on it's always such a pleasure to to learn more about the work they're doing in terms of what you've got coming up OB so this paper that is coming out in the future uh that's out people can read it although it's supposedly out in the future um what else what else have you got coming up where can people find your work how can people reach out to you if they're interested in asking questions or collaborating or whatever it might be oh yeah I mean you know any anything on our you know our lab website uh you know or email me you know like R rsmith lur institute.org you know feel free uh you know we'll link the L Twitter I mean you know any any know the typical Roots all the all the standard the standard stuff I mean my email is also typically the one that's like the corresponding author email on most papers so um you know there's certainly Lots uh lots of ways to get in contact me with you so desire but great well we'll make sure we link everything below uh so thank you once again from me and The Institute it's been really really fun sure no thank you very much for having me